

EVALPAX: Evaluation of a pharmacist-driven protocol for nirmatrelvir/ritonavir prescribing in a community hospital system

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Purpose: The purpose of this study was to evaluate pharmacist prescribing of nirmatrelvir/ritonavir to ensure this method of increasing access to treatment is safe and effective.

Methods: This multicenter, retrospective observational study included patients receiving a prescription for nirmatrelvir/ritonavir by a physician, nurse practitioner (NP), physician assistant (PA), or pharmacist at an Indiana University (IU) Health West Central Region site over a 3-month period. Patients were divided into two groups: those who received nirmatrelvir/ritonavir prescribed by a pharmacist (the pharmacist prescribed group) and those who received nirmatrelvir/ritonavir prescribed by other providers (the physician/NP/PA prescribed group). Electronic health record (EHR) reviews were performed to assess the appropriateness of prescriptions based on the presence of risk factors and symptoms, day of symptom onset, and dosing. The primary endpoint was the overall appropriateness of nirmatrelvir/ritonavir prescriptions in the two study groups based on emergency use authorization inclusion and exclusion requirements. Secondary endpoints included appropriateness of nirmatrelvir/ritonavir dosing and medically attended visits or mortality within 30 days. Statistical analysis of the endpoints occurred post hoc utilizing the Fisher's exact test.

Results: A total of 259 patients were included in the pharmacist prescribed group and 265 patients in the physician/NP/PA prescribed group. Overall appropriate nirmatrelvir/ritonavir prescribing occurred in 258 patients (99.6%) and 232 patients (87.5%) in the pharmacist and physician/NP/PA prescribed groups, respectively ($P < 0.0001$). Nirmatrelvir/ritonavir dosing was appropriate in 256 patients (98.8%) and 240 patients (90.6%) in the pharmacist and physician/NP/PA prescribed groups, respectively ($P < 0.0001$). The 30-day rates of medically attended visits were similar between groups. No patients died within 30 days of treatment in either group.

Conclusion: Pharmacist prescribing of nirmatrelvir/ritonavir may result in a higher likelihood of prescriptions meeting overall appropriateness criteria. Pharmacists represent an important healthcare professional resource to improve nirmatrelvir/ritonavir prescribing and utilization.

Keywords: COVID-19, nirmatrelvir, pharmacist prescribing, quality improvement, ritonavir

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The coronavirus disease 2019 (COVID-19) pandemic introduced a multitude of challenges to healthcare providers and systems around the world, one of which included identifying an effective antiviral treatment option.¹ Antiviral treatment for hospitalized

patients became available quickly in the United States with the remdesivir emergency use authorization (EUA) in May 2020.² Outpatient-focused antiviral treatment lagged behind, with the first treatment option, bamlanivimab, becoming available in November 2020.³ Additional

monoclonal antibodies were developed over the course of the pandemic, but each has suffered from a short lifespan due to SARS-CoV-2 variant formation resulting in monoclonal antibody resistance.⁴ Oral treatment options have also been developed, including nirmatrelvir/ritonavir and molnupiravir, which were approved in December 2021.^{5,6} Both options have remained active against circulating variants and continue to be recommended for use in the National Institutes of Health COVID-19 treatment guidelines.^{7,8}

When outpatient-focused COVID-19 treatment options were first authorized, the Indiana University (IU) Health system created a centralized referral center staffed by physicians, nurse practitioners (NPs), physician assistants (PAs), and pharmacists to coordinate outpatient treatment. This approach was adopted because monoclonal antibodies infusions were initially used for patients at the highest risk for COVID-19 progression (eg, the immunocompromised) and required the scheduling of appointments. Although the referral center approach standardized and increased access to outpatient COVID-19 treatment, many challenges remained, including monoclonal antibody availability and logistical constraints.⁹

With the EUA for oral nirmatrelvir/ritonavir (Paxlovid), the access to outpatient COVID-19 treatment was thought to be greatly expanded.⁵ However, nirmatrelvir/ritonavir requires several considerations prior to prescribing, most notably a review of a patient's renal function and home medication list to identify possible drug interactions.⁷ At IU Health, these concerns prevented many outpatient physicians/NPs/PAs from feeling comfortable prescribing this medication. The referral center began evaluating patients for nirmatrelvir/ritonavir treatment, but delays in care remained due to the overwhelming number of referrals and steps needed to complete the EUA requirements.

In July 2022, the Food and Drug Administration (FDA) authorized pharmacist prescribing of nirmatrelvir/

KEY POINTS

- Pharmacists were permitted to prescribe nirmatrelvir/ritonavir according to the emergency use authorization to expand access to COVID-19 treatment but with significant limitations related to drug-drug interactions.
- Indiana University Health developed a protocol to expand pharmacist prescribing of nirmatrelvir/ritonavir by allowing minor home medication dose adjustments without physician or other provider approval.
- The pharmacist prescribing protocol led to patients having a higher likelihood of receiving an appropriate prescription for nirmatrelvir/ritonavir than those in a comparison group whose nirmatrelvir/ritonavir was prescribed by a physician, nurse practitioner, or physician assistant, although confirmation that patients picked up prescriptions was not performed.

ritonavir as another step to increase access to outpatient COVID-19 treatment.⁵ Pharmacists are uniquely trained and accessible, making them an optimal resource for delivery of nirmatrelvir/ritonavir.¹⁰ However, under the July 2022 authorization pharmacists were limited in their prescribing. Pharmacists were not permitted to prescribe nirmatrelvir/ritonavir in cases where there was insufficient information regarding a patient's renal and hepatic function, there was insufficient information on their use of other medications for drug interaction verification, and/or modification of one of a patient's other medication regimens was required.¹¹ Due to the extensive drug interactions caused by nirmatrelvir/ritonavir, pharmacist

prescribing was limited in many settings. For example, in IU Health's West Central Region (a regional health system within the larger IU Health system), community pharmacists prescribed nirmatrelvir/ritonavir from August to December 2022, but only a total of 43 patients were able to receive pharmacist-prescribed nirmatrelvir/ritonavir in this 4-month span due to these limitations.

In December 2022, the IU Health system developed a protocol to allow ambulatory care clinical pharmacists to prescribe nirmatrelvir/ritonavir, including the ability to make minor dose adjustments to other medications. For example, a pharmacist could instruct a patient to reduce a dose or hold a medication such as a statin or antihypertensive during treatment with nirmatrelvir/ritonavir. The goal of this protocol was to increase accessibility to COVID-19 treatment, standardize the triage process, improve interaction management, and reduce physician/NP/PA burden. Similar protocols have been evaluated previously, and various approaches have been described, including physician/NP/PA, patient, or pharmacist initiation of the protocol.¹²⁻¹⁴

At IU Health, patients called a triage telephone line staffed by nurses to report a positive COVID-19 result, to answer a series of standardized questions to determine if they met EUA-defined criteria for treatment with nirmatrelvir/ritonavir, and to confirm their current medication list. Nurses discussed these topics with every patient who contacted the triage line, and then sent messages through the electronic health record (EHR) to a group of ambulatory care clinical pharmacists. Messages were only sent to the ambulatory care clinical pharmacists and did not involve the patient's primary care physician/NP/PA. These pharmacists would in turn evaluate the appropriateness of therapy, manage drug interactions, and send nirmatrelvir/ritonavir prescriptions to a patient's preferred pharmacy. If the ambulatory care clinical pharmacist deemed nirmatrelvir/ritonavir

therapy inappropriate due to drug interactions, a referral for monoclonal antibody therapy was placed to the centralized referral center. Lastly, the ambulatory care clinical pharmacist contacted each patient or patient caregiver to review the decision-making and treatment plan moving forward.

This study was conducted within the West Central Region of the IU Health system. The purpose of the study was to evaluate pharmacist prescribing of nirmatrelvir/ritonavir pursuant to this protocol, as compared to prescribing by physicians/NPs/Pas, to ensure this approach is safe for patients and is associated with similar clinical outcomes.

Methods

EVALPAX was a multicenter, retrospective observational study to evaluate nirmatrelvir/ritonavir prescribing at any of the 3 hospitals in the IU Health West Region (IU Health Arnett, White Memorial, and Frankfort hospitals) and their associated outpatient clinics. IU Health Arnett is a 191-bed, community teach hospital, while IU Health White Memorial and Frankfort hospitals are critical access hospitals with less than 30 beds at each location. This study was approved by the IU institutional review board (IRB-19992).

Patients were included in the study if they were prescribed nirmatrelvir/ritonavir by an ambulatory care clinical pharmacist or physician/NP/PA from December 15, 2022, through March 31, 2023. This approximately 3-month period was selected as it was during the entire time the pharmacist prescribing protocol was utilized. The protocol ended in April 2023. The ambulatory care clinical pharmacists varied in experience and level of training, though all received training related to nirmatrelvir/ritonavir prescribing and considerations. Patients were divided into two groups: those who received nirmatrelvir/ritonavir prescribed by a pharmacist (the pharmacist prescribed group) and those who received nirmatrelvir/ritonavir prescribed by other providers (the physician/NP/PA prescribed group). For both groups,

patients had their EHR reviewed retrospectively by investigators to collect data including age, sex, risk factors for COVID-19 progression, day of symptom onset reported by the patient to the nurse, nirmatrelvir/ritonavir dosing strategy, overall appropriateness of nirmatrelvir/ritonavir prescribing, reason for inappropriateness of nirmatrelvir/ritonavir (if relevant), and medically attended visits and mortality within 30 days of the prescription.

Risk factors for COVID-19 progression were based on the Centers for Disease Control and Prevention list of underlying medical conditions associated with a higher risk of severe COVID-19.¹⁵ Risk factors have evolved over the course of the pandemic. Risk factors included in this study were age greater than 64 years, heart disease, chronic lung disease, chronic kidney disease, asthma, cystic fibrosis, active cancer treatment, cerebrovascular disease, type 1 or 2 diabetes mellitus, body mass index greater than 30 kg/m², physical disability, mental health condition, neurologic disease, history of or current smoking, pregnancy, human immunodeficiency virus (HIV) infection, chronic systemic steroid use, solid organ transplantation, hematopoietic stem cell transplantation, and other, unspecified conditions.

For the pharmacist prescribed group, data was collected on the pharmacist decision-making process. Pursuant to the protocol, a pharmacist reviewed a patient's chart to determine if nirmatrelvir/ritonavir could be safely prescribed and documented the decision in the EHR. The ambulatory care clinical pharmacists utilized an internal drug interaction reference based on the nirmatrelvir/ritonavir EUA document, the Liverpool COVID-19 Interaction Checker, and Lexicomp Drug Interactions app (UpToDate Inc., Waltham, MA). The internal reference was reviewed by pharmacists and physicians to determine which drug interactions were due to contraindications and which were not, given the nuances of many drug interactions. Examples of drug interactions for

which a pharmacist was allowed to make adjustments (eg, decrease dose or hold medication) included those involving statins and antihypertensives. Examples of contraindicated drug interactions include anticoagulation and immunosuppression. If a patient required laboratory monitoring (ie, renal function assessment) prior to prescribing of nirmatrelvir/ritonavir, the pharmacist messaged the physician or advanced practice provider (APP) to order the required laboratory test, and the decision to prescribe nirmatrelvir/ritonavir was left to the physician/APP because this protocol was in addition to the pharmacists' other job responsibilities and they were not able to follow patients over several days to ensure the laboratory results were obtained and evaluated. Table 1 illustrates the decision-making process.

The primary outcome of this study was the overall appropriateness of nirmatrelvir/ritonavir prescribing, which was assessed on the basis of meeting EUA inclusion and exclusion requirements. Inappropriate use criteria included the following: the patient was asymptomatic, symptom onset was more than 5 days before nirmatrelvir/ritonavir initiation, no risk factors for disease progression were present, there was a contraindicated drug-drug interaction and the patient still received nirmatrelvir/ritonavir, the estimated glomerular filtration rate (eGFR) was less than 30 mL/min/1.73m², and no renal function result obtained within the last year was available.

Secondary outcomes included the appropriateness of nirmatrelvir/ritonavir dosing, a medically attended visit within 30 days of prescription, and mortality within 30 days of prescription. A medically attended visit was defined as an emergency department encounter, urgent care encounter, or hospital admission. Mortality within 30 days of prescription was evaluated using EHR records only.

Statistical analysis was performed post hoc using SAS version 9.4 (SAS Institute, Cary, NC). Descriptive statistics were used for baseline

Table 1. Pharmacist Decision-Making Process

Pharmacist decision	Explanation of decision	Pharmacists primary action	Pharmacist secondary action
Prescribe nirmatrelvir/ritonavir without modifications to home medication regimen	Pharmacist identified no drug-drug interactions nor renal insufficiency	Send prescription to patient's pharmacy	None
Prescribe nirmatrelvir/ritonavir with modifications to home medication regimen or renal adjustment	Pharmacist identified minor drug-drug interactions or necessary renal dosing adjustment with prescribing of nirmatrelvir/ritonavir	Send prescription to patient's pharmacy; patient instructed by pharmacist to make minor modification to home medication regimen	None
Contraindication or contraindicated drug-drug interaction identified	Pharmacist identified eGFR <30 mL/min or a drug-drug interaction resulting in a contraindication to prescribing nirmatrelvir/ritonavir	Send referral for monoclonal antibody or remdesivir treatment	Notify physician/NP/PA of contraindication and referral for alternative treatment options
Complex drug-drug interaction identified	Pharmacist identified drug-drug interactions that were complex in nature, requiring monitoring while on treatment or significant changes to a patient's home medication regimen	Pharmacist does not prescribe nirmatrelvir/ritonavir	Pharmacist provides recommendations to physician/NP/PA regarding the complex drug-drug interactions; physician/NP/PA may choose to prescribe based on their own discretion
Nirmatrelvir/ritonavir is inappropriate	Pharmacist identified no indication for nirmatrelvir/ritonavir treatment due to patient being outside symptom onset window or not having any risk factors for disease progression	Pharmacist does not prescribe nirmatrelvir/ritonavir	Notify physician/NP/PA of patient being ineligible for nirmatrelvir/ritonavir treatment
Laboratory results needed before prescribing	Patient has not had renal function monitoring within the last year	Pharmacist does not prescribe nirmatrelvir/ritonavir	Notify primary care physician/NP/PA of the required laboratory results

Abbreviations: eGFR, estimated glomerular filtration rate; NP, nurse practitioner; PA, physician assistant.

characteristics. The Fisher's exact test was utilized to evaluate the primary and secondary endpoints as they were categorical. To determine statistical significance, a 2-tailed α of 0.05 was used. A power calculation was not performed, as the sample of patients evaluated was a convenience sample.

Results

A total of 682 patients were evaluated for inclusion in the study, which resulted in 524 patients being included. In the pharmacist prescribed group, 259 of 395 evaluated patients were included, while 265 of 287 evaluated patients were included in the physician/NP/PA prescribed group. The most common reasons for exclusion from the pharmacist prescribed group were the presence of a contraindicated

drug-drug interaction and that the patient was not an appropriate candidate for treatment or required laboratory results prior to treatment. If a patient was excluded from the pharmacist prescribed group and still received a prescription for nirmatrelvir/ritonavir, they were included in the physician/NP/PA group. In the physician/NP/PA group, patients were most commonly excluded due to being referred to the centralized referral center for treatment. Of the 259 patients in the pharmacist prescribed group, 152 patients (58.7%) were prescribed nirmatrelvir/ritonavir after the pharmacists made adjustments to home medications.

Baseline characteristics of the groups were comparable (Table 2). The median age of the overall population was 64 years (interquartile range, 49-73

years). The median day of symptom onset at the time of treatment initiation was 2 days in both groups, and approximately 81% of the overall patient population had received at least one dose of a COVID-19 vaccine at the time of treatment. The most common risk factors for COVID-19 disease progression included age greater than 65 years (47.7%), BMI greater than 30 kg/m² (47.1%), mental health condition (27.7%), and type 1 or 2 diabetes mellitus (21.2%).

For the primary outcome of appropriateness of nirmatrelvir/ritonavir prescribing, 258 of 259 patients (99.6%) in the pharmacist prescribed group and 232 of 265 patients (87.5%) in the physician/NP/PA prescribed group received a prescription within EUA requirements ($P < 0.0001$). For inappropriately

Table 2. Baseline Characteristics by Study Group

Characteristic	Pharmacist prescribed group (n = 259)	Physician/NP/PA prescribed group (n = 265)	Total (N = 524)
Age, years			
Median (IQR)	66 (52-73)	61 (45-72)	64 (49-73)
Range	24-95	17-94	17-95
Sex, No. (%)			
Male	81 (31)	100 (38)	181 (35)
Female	178 (69)	165 (62)	343 (65)
Time since symptom onset when treatment initiated, days			
Median (IQR)	2 (1-3)	2 (2-3)	2 (1-3)
Range	1-5	1-11	1-11
Vaccination status, No. (%)			
Unvaccinated	32 (12.4)	61 (23)	93 (17.7)
1 vaccine dose	4 (1.5)	8 (3)	12 (2.3)
2 vaccine doses	33 (12.7)	39 (14.7)	72 (13.7)
3 vaccine doses	74 (28.6)	65 (24.5)	139 (26.5)
>3 vaccine doses	114 (44)	87 (32.8)	201 (38.4)
Not documented	2 (0.8)	5 (1.9)	7 (1.4)
Risk factors, No. (%)			
Age >65 years	139 (53.7)	111 (42)	250 (47.7)
BMI >30 kg/m ²	126 (48.6)	121 (45.8)	247 (47.1)
Mental health condition	79 (30.5)	66 (25)	145 (27.7)
Type 1 or 2 diabetes mellitus	55 (21.2)	56 (21.2)	111 (21.2)
Chronic kidney disease	33 (12.7)	32 (12.1)	65 (12.4)
Smoking (current or history)	24 (9.3)	29 (11)	53 (10.1)
Heart disease	26 (10)	34 (12.9)	60 (11.5)
Asthma	32 (12.4)	14 (5.3)	46 (8.8)
Chronic lung disease	19 (7.3)	14 (5.3)	33 (6.3)
Immunocompromised	8 (3.1)	2 (0.8)	10 (1.9)
Active treatment for cancer	3 (1.2)	5 (1.9)	8 (1.5)
Cerebrovascular disease	2 (0.8)	3 (1.1)	5 (1)
Neurologic disease	1 (0.4)	0	1 (0.2)
Physical disability	0	1 (0.4)	1 (0.2)
Pregnancy	0	1 (0.4)	1 (0.2)
Other risk factor ^a	47 (18.1)	2 (0.8)	49 (9.4)

Abbreviations: IQR, interquartile range; NP, nurse practitioner; PA, physician assistant.

^aOther risk factors included those with mixed evidence suggesting a higher risk of COVID-19 (eg, hypertension).

prescribed nirmatrelvir/ritonavir, 1 patient in the pharmacist prescribed group had an eGFR less than 30 mL/

min/1.73m². Thirty-three patients in the physician/NP/PA prescribed group were prescribed nirmatrelvir/ritonavir

not in accordance with EUA requirements. The most common reasons were no risk factors for severe COVID-19

being present, symptom onset outside the 5-day window, and no laboratory results being available for assessment (Table 3).

Appropriateness of nirmatrelvir/ritonavir dosing was assessed as a secondary outcome. For appropriate dosing, 256 of 259 patients (98.8%) in the pharmacist prescribed group received an appropriate dose, while 240 of 265 (90.6%) of patients in the physician/NP/PA prescribed group received an appropriate dose ($P < 0.0001$). The occurrence of medically attended visits within 30 days of treatment was comparable between the two groups, with 18 patients (6.9%) and 16 patients (6%) having a visit in the pharmacist prescribed and physician/NP/PA prescribed groups, respectively. No patients in either group died within 30 days of treatment (Table 3).

Discussion

In this retrospective review, patients had a higher likelihood of receiving an appropriately prescribed and dosed nirmatrelvir/ritonavir prescription from a clinical pharmacist than from another type of healthcare provider. Clinical outcomes such as medically attended visits and mortality were similar in the two study groups, suggesting pharmacists may be able to safely and

effectively prescribe nirmatrelvir/ritonavir for the outpatient treatment of COVID-19.

The COVID-19 pandemic was overwhelming for all kinds of healthcare professionals, leading to work overload and burnout.¹⁶ Physicians and other healthcare providers were especially burdened during the COVID-19 pandemic by the many responsibilities added to an already stressful work environment. In May 2022, the US Department of Health and Human Services' assistant secretary for planning and evaluation released an issue brief detailing these many stressors and overwhelming situations experienced by healthcare providers.¹⁷ Although the authorization of nirmatrelvir/ritonavir introduced the first oral, more easily accessible treatment option for outpatient COVID-19 treatment, its authorization came with many considerations and frustrations for all providers, including renal dose adjustments and a multitude of complex drug-drug interactions. A 2023 Medscape survey polling more than 1,500 providers revealed the top 4 reasons for not prescribing nirmatrelvir/ritonavir were related to these two considerations.¹⁸ Additionally, nirmatrelvir/ritonavir is associated with unpleasant adverse effects including diarrhea, dysgeusia,

and myalgias.^{19,20} This has led to sub-optimal acceptance of prescribing by providers.²¹

Pharmacists represent an invaluable resource for scenarios such as nirmatrelvir/ritonavir prescribing. Pharmacists are specially trained to evaluate drug-drug interactions and adjust medication doses based on renal function. Furthermore, pharmacists can counsel patients on expected adverse effects and instruct patients on how to manage these effects or when to discontinue the medication. Nirmatrelvir/ritonavir represents a unique opportunity to expand pharmacist prescribing authority, with appropriate limitations. As outlined in the IU Health protocol, pharmacists were able to prescribe nirmatrelvir/ritonavir for patients with or without minor drug-drug interactions and make renal dose adjustments. This allowed many patients to be evaluated by a pharmacist and receive a prescription without needing input from a physician/NP/PA, thus providing some relief for physicians/NPs/PAs with an overwhelming number of treatment requests. Due to the general straightforward approach to nirmatrelvir/ritonavir prescribing, which includes documenting a positive COVID-19 test, confirming symptom onset within 5 days, and evaluating for

Table 3. Comparison of Nirmatrelvir/Ritonavir Prescribing Practices

Outcome	Pharmacist prescribed group (n = 259)	Physician/NP/PA prescribed group (n = 265)	P value
Overall appropriateness of nirmatrelvir/ritonavir prescription, No. (%)	258 (99.6)	232 (87.5)	<0.0001
Reasons for an inappropriate nirmatrelvir/ritonavir prescription, No. (%)			
No risk factors present	0	20	
Symptom onset >5 days ago	0	8	
No laboratory tests within 1 year	0	4	
Contraindicated drug-drug interaction	0	1	
eGFR <30 mL/min/1.73m ²	1	0	
Appropriateness of nirmatrelvir/ritonavir dosing, No. (%)	256 (98.8)	240 (90.6)	<0.001
Medically attended visit within 30 days, No. (%)	18 (6.9)	16 (6)	0.68
Mortality within 30 days, No. (%)	0	0	

Abbreviations: eGFR, estimated glomerular filtration rate; NP, nurse practitioner; PA, physician assistant.

drug-drug interactions and renal dose adjustments, pharmacists can review and interpret this information with little ambiguity. Another highlight of this study is that over half of the pharmacist prescribing instances involved pharmacist modification of patients' home medication regimens. This emphasizes that in addition to a pharmacist being able to manage evaluating a patient for treatment, pharmacists are also able to make accurate adjustments to home medication regimens to safely account for drug interactions. However, as in the IU Health protocol, certain limitations must remain in place for challenging situations due to either drug-drug interactions or other clinical scenarios. As a result, pharmacists were required to discuss complicated drug-drug interactions (eg, if multiple drug interactions were present or significant monitoring was required) with a physician/NP/PA prior to the patient receiving treatment to ensure patient safety.

Pharmacist prescribing is not a new practice and has gained popularity in recent years. For example, pharmacists are permitted to prescribe naloxone across the US, while many states have started to allow further expansion of pharmacist prescribing to oral contraceptives and smoking cessation products.²² Pharmacist prescribing of nirmatrelvir/ritonavir represents another expansion of this prescribing authority to assist physicians/NPs/PAs in patient care. Additionally, pharmacist involvement with nirmatrelvir/ritonavir has been evaluated previously. A recent study from the VA Butler Healthcare System described a pharmacy consult service allowing pharmacists to prescribe nirmatrelvir/ritonavir.¹⁴ Physicians/NPs/PAs assessed patient eligibility and placed a consult for pharmacy review if a patient met eligibility criteria. The IU Health protocol was different in terms of protocol initiation as it enabled a nurse to be the starting point after discussion with a patient. This approach was adopted to further reduce the time burden on physicians, NPs, and PAs.

A second study conducted in a New Brunswick health network described a protocol similar to the IU Health protocol, with the involvement of nurses and pharmacists, though results were only reported on the number of prescriptions sent to pharmacies. Lastly, a study from University of Wisconsin Health described a pharmacist-driven protocol.¹² This study evaluated a smaller number of patients and did not compare pharmacist prescribing practices to those of physicians/APPs, but the study did confirm if patients picked up the nirmatrelvir/ritonavir prescription, which was not done in our study. Our study adds to this growing data by including a larger number of patients and comparing prescribing between pharmacists and physicians/NPs/PAs.

This study has notable strengths and limitations. The multicenter design and number of patients included helps generalize the results to community health systems. However, this was a retrospective, observational study, thus introducing the possibility of selection bias. The authors were also reliant upon proper documentation in the EHR to collect data, which could have introduced confounding, especially considering outcome identification was only noted if it occurred within the IU Health system. Prescriptions for nirmatrelvir/ritonavir were identified using the EHR. The authors did not confirm whether the medication was picked up from the pharmacy and taken for the full duration. Lastly, the authors reviewed the EHR independently, and audits between authors were not performed.

Conclusion

Pharmacist prescribing of nirmatrelvir/ritonavir in a community health system setting may result in a higher likelihood of patients receiving a prescription meeting EUA requirements and at an appropriate dose, with similar clinical outcomes, relative to a nirmatrelvir/ritonavir prescribing by a physician, NP, or PA. Pharmacists represent an important group of healthcare professionals in efforts to improve

nirmatrelvir/ritonavir prescribing and utilization.

Disclosures

The authors have declared no potential conflicts of interest.

Additional information

The authors provided the following information regarding their contributions to this work. All authors made substantial contributions to the design and completion of this work. All authors were involved in drafting and revising the work. All authors agree to be accountable for all aspects of the work.

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